

To: Lab #1
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Subject: EME 109

Opening

The purpose of this memo is to present the result of an experiment in which measured viscous fluid flow effects around a circular cylinder in a open-circuit wind tunnel. We were tasked with determining the surface pressure profiles, observing boundary layer behavior, and computing the total drag force exerted on the cylinder. In this report, we outline our findings and compare them to established measurements.

Finding

The key findings of the experiment include the derivation of the experimental pressure drag force D_p which was found to be approximately 4.8525 N via numerical integration of the surface pressure data. This aligned closely with available published data that predicted a drag force of 5.4985 N, resulting in a minor error of 11.75%. In addition, we determined that boundary layer separation occurred at an angle of approximately 105°, indicating a turbulent boundary layer before separation.

Discussion

Methods

The testing apparatus involved placing a circular cylinder within the test section of an open-circuit wind tunnel. At a free-stream airspeed of 20 m/s as set by the TA, we slowly rotated the cylinder from 0° to 180° in 10° increments. At each of the intervals, we recorded the surface gauge pressure using a surface pressure tap in addition to the free-stream dynamic pressure recorded by a Pitot-static probe and system drag recorded by a mechanical force balance system. We also performed additional evaluations using the Pitot-static probe to determine the velocity profile up-stream.

Analysis

We began analysis by adjusting the standard air density for the localized laboratory conditions [1], those being 21°C and 43% humidity respectively. The revealed a new density of ρ equals 1.2 kg/m³. We then converted the cylinder manometer heights to gauge surface pressures in pascals. We determined the experimental pressure coefficient across all tap angles by normalizing the surface gauge pressure against the dynamic pressure obtained from the Pitot-static probe. We then applied a trapezoidal Riemann sum to numerically integrate our surface pressure data over the geometry of the cylinder to calculate the net pressure drag force (D_p). Lastly, we used White's published drag coefficient of 1.2 for an infinite cylinder [2] to calculate our theoretical total drag force (F_d).

Results

We observed that the surface pressure reached its maximum positive peak at the 0° stagnation point, dropped to its lowest negative peak of -0.6335 kPa at 80°, and then leveled off into a stable

plateau between 130° and 180°. When we plotted our experimental C_p alongside inviscid flow theory ($C_p = 1 - 4 \sin^2 \theta$), we found that the two models closely aligned until the flow moved past 80°. In our physical results, the pressure coefficient never fully recovered to 1.0 at the rear of the cylinder, remaining stuck near -1.0 instead. Our calculations yielded an experimental pressure drag of 4.8525 N, which sits right below our calculated theoretical drag force of 5.4985 N and our average measured force balance drag of 4.29 N.

Commentary

We discuss in Attachment III the stark differences between our experimental curve and the inviscid flow curve, which we believe to be the result of fluid friction and viscosity. Because inviscid theory completely ignores friction, it assumes the fluid can recover its full pressure behind the body. In our actual test, the pressure gradient past 80° forces the flow to separate. While a laminar boundary layer separates at 82°, our flow profile was closer to the turbulent model, which doesn't separate until 105°. We discovered the reason for this early turbulence when analyzing our lateral traverse data in Attachment IX; the plot showed that airspeed dropped by over 1 m/s across the center of the tunnel, proving that we did not have a uniform velocity profile upstream of the cylinder.

Closing

In conclusion, we demonstrated how viscosity and boundary layer separation dominate the flow profile and net drag forces acting on a cylindrical body. Our lateral traverse data proved that our wind tunnel facility did not provide a completely uniform free-stream velocity profile upstream. We recommend that future laboratory sessions optimize the upstream flow-straightening grids to eliminate these velocity fluctuations before testing. For further reference, please see the references and attachments below

References

- [1] D. Czernia and B. Szyk, "Air Density Calculator."
- [2] F. M. White, *Fluid mechanics*, 6. ed., 3. print. in McGraw-Hill series in mechanical engineering. Boston: McGraw-Hill Higher Education, 2008.

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Attachment I - Adjusted Air Density

While the density of air is given at 1.225 kg/m³, we must adjust it for humidity and temperature. This can be done using partial pressure analysis. For this lab, I have opted to use an online tool [1], which at a temperature of 21 C° and 43% humidity gives a air density of 1.200764 kg/m³.

Attachment II - Pressure vs. Angle

From the lab, we recorded the following data.

Prssure Tap Angle (degrees)	Cylinder Manometer (kPa)	Airspeed Manometer (mmH2O)	Measured Drag (N)
0	0.2972	28.0	4.30
10	0.2835	26.5	4.22
20	0.2030	26.5	4.33
30	0.0695	26.5	4.22
40	-0.1014	26.0	4.10
50	-0.2830	26.0	4.15
60	-0.4435	26.0	4.22
70	-0.5834	26.0	4.16
80	-0.6335	26.0	4.18
90	-0.5846	26.0	4.23
100	-0.5610	26.0	4.35
110	-0.3757	26.5	4.37
120	-0.3091	26.5	4.27
130	-0.2696	26.5	4.31
140	-0.2558	26.5	4.29
150	-0.2533	26.5	4.32
160	-0.2674	26.5	4.43
170	-0.2358	26.5	4.45
180	-0.2335	26.5	4.37

Table 1: Tabulated data recorded from cylinder experiment recorded during lab session.

To be able to plot the manometer pressures in Pascals against the tap angle, we need to convert the units of the cylinder manometer by multiplying by 1000. To get the airspeed manometer into Pa, we need to use the following equation to turn the height of water into pressure.

$$P = \rho_{\text{water}} \cdot g \cdot h_{\text{water}}$$

Plotting this, we get Figure 1.

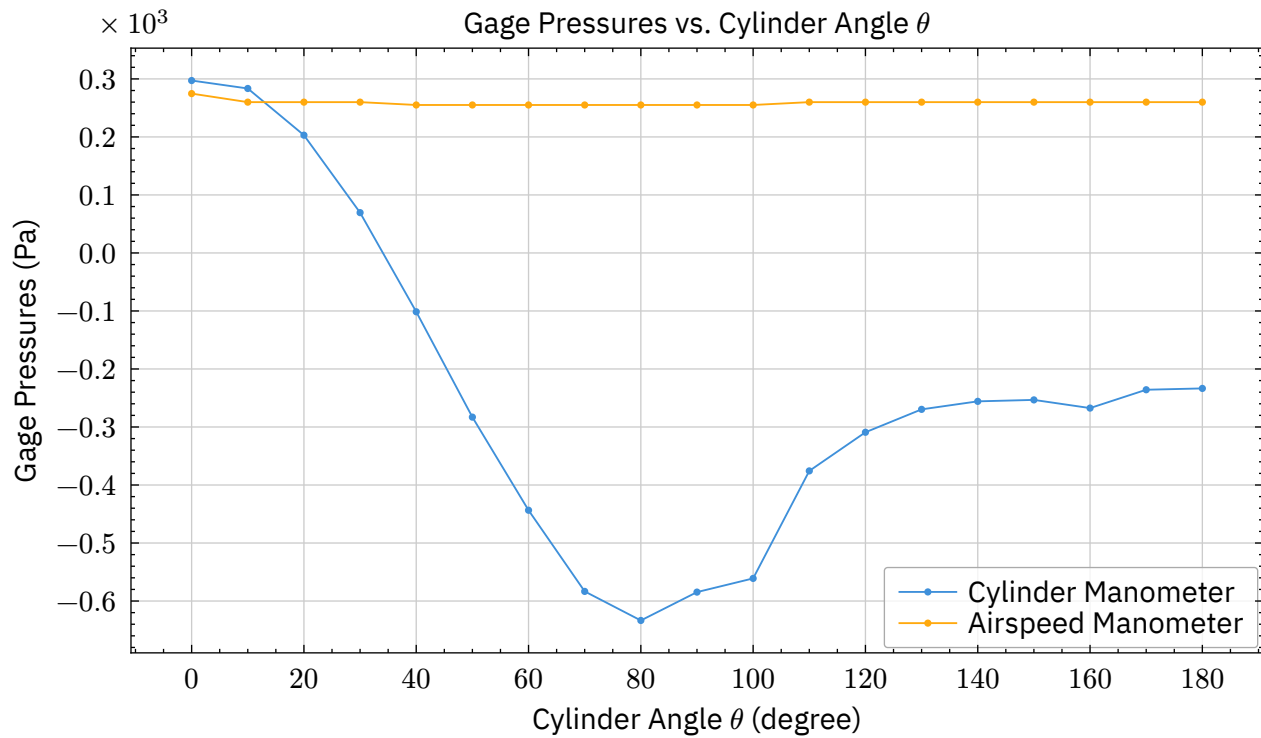


Figure 1: Plot of the measured gauge pressures as a function of the angle θ on the cylinder surface.

Attachment III - Pressure Coefficient vs. Angle

To find the experimental pressure coefficient C_p , we use the equation provided to us in the lab manual.

$$C_p = \frac{p_s - p_\infty}{\frac{1}{2}\rho V^2}$$

The recorded data from the lab already gives us the manometer values in gauge pressure, so we can plug it into the numerator as is. Similarly, the denominator is simply the expanded form of the dynamic pressure which we recorded with the Pitot-static tube. We simplify the equation to the following:

$$C_p = \frac{P_{\text{cylinder surface}}}{P_{\text{dynamic}}}$$

We compare this with the inviscid flow C_p as defined by the following equation:

$$C_p = 1 - 4 \sin^2 \theta$$

Running the data, we get Figure 2.

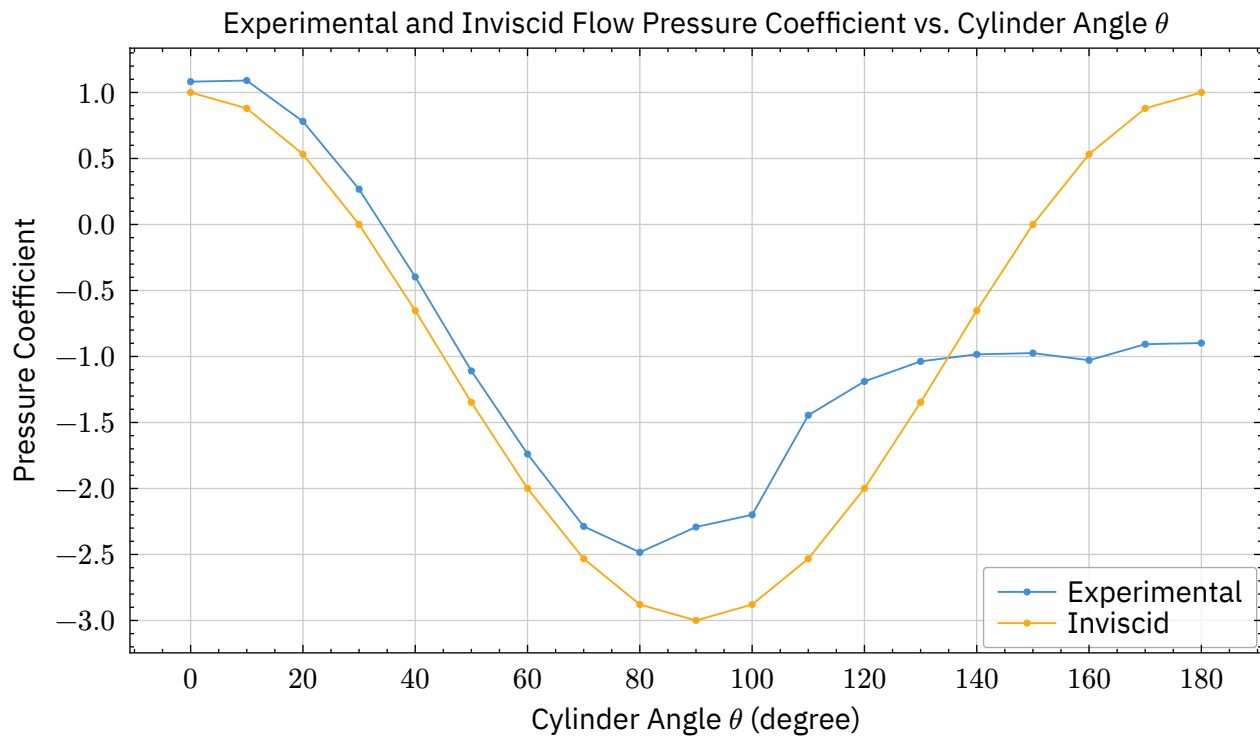


Figure 2: Plot of the experimental and inviscid flow pressure coefficients as a function of the angle θ on the cylinder surface.

Comparing the experimental and inviscid coefficients, we can see that they are closely aligned until after 80° , where the experimental data peaks earlier before leveling off at about $C_p = -1.0$. The inviscid flow by definition assumes that there is no friction in the flow, hence why the coefficient is able to recover fully after moving about the cylinder. The presence of friction in the experimental creates turbulence after the wake, decreasing the speed of the flow and increasing its pressure. This is what causes the difference in pressure coefficients.

Attachment IV - Finding Pressure Drag

The lab manual gives us the following equation to calculate the pressure drag.

$$D_p = 2RL \int_0^\pi (p_s - p_\infty) \cos \theta d\theta$$

To compute this numerically, we must perform a trapezoidal remains sum represented with the following equation.

$$D_p \approx 2RL \sum_{i=1}^{n-1} \left(\frac{\Delta P_i \cos \theta_i + \Delta P_{i+1} \cos \theta_{i+1}}{2} \right) \Delta \theta_i$$

Implementing this in code (Attachment X) and solving, we find that the **resulting drag force is approximately 4.8525 N.**

Attachment V - Published Pressure Drag

Based on the fluid mechanics textbook by White [2] Table 7-3, if we approximate the cylinder as having an infinite length to diameter ratio, the drag coefficient for the body is 1.2.

We can use this drag coefficient and the following equation to find the equivalent drag force on the cylinder, where the impacted surface is the cross-sectional area of the cylinder $A = 2RL$.

$$F_d = C_d \cdot \frac{1}{2} \rho_{\text{air}} u_{\infty}^2 \cdot A$$

Using the proper quantities, we find the **theoretical drag force F_d is 5.4985 N.**

Attachment VI - Comparing Drag Force

The experimental and theoretical values are quite close to each other, only exhibiting a 11.75% error in value. The value in the textbook assumes completely laminar flow. As we will discuss in Attachment VII, the flow in the experiment is not completely laminar. The increased turbulence would reduce the total drag the the cylinder experiences, thus reducing the drag coefficient.

Attachment VII - Flow Separation

Once again, we can reference the fluid mechanics book from White [2] to find the following figure, which showcases the types of flow separation around a cylinder.

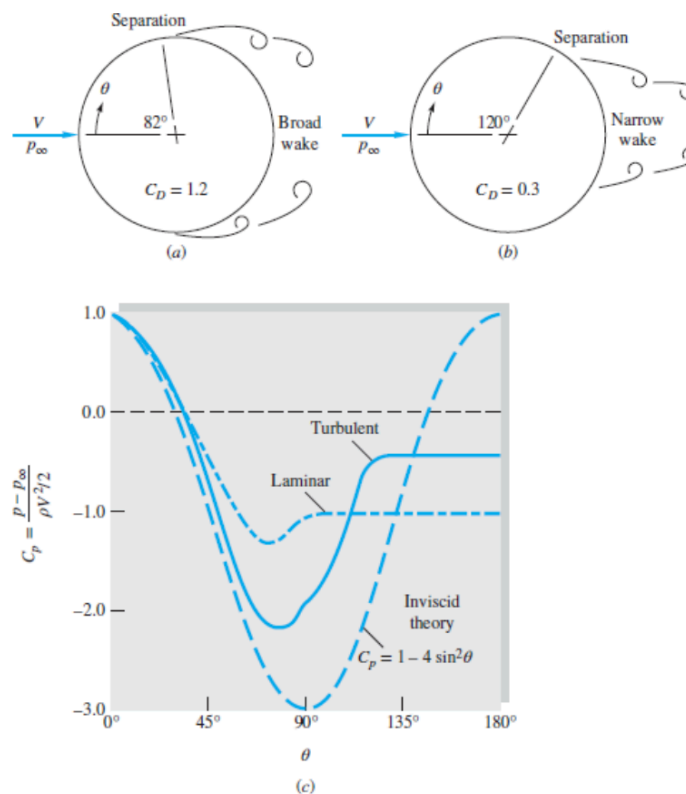


Figure 3: Flow separation diagram from White [2].

Our experiment began experiencing flow separation at around 105°. Comparing this with our data from Figure 2, we find that our flow follows a trend closer to the turbulent model which suggests that flow separation should occur at 120° where as laminar flow separation should occur 82°.

While the apparatus is designed to create as laminar a flow as possible, we note in Attachment IX that the flow entering the apparatus is not uniform. This implies the presence of turbulence in the system.

Attachment VIII - Pitot-Static Probe Angle Effects

In lab, we collected the following data:

Probe Angle (degrees)	Manometer Reading (mmH2O)
20	28
10	28.5
0	29.5
-10	28.5
-20	27

Table 2: Tabulated data of the pitot-static probe as it is rotated.

We can rearrange the pressure equation to get apparent air speed.

$$\Delta P = \frac{1}{2}\rho_{\infty}u_{\infty}^2 = \rho gh \Rightarrow u_{\infty} = \sqrt{\frac{2\rho gh}{\rho_{\infty}}}$$

With this, we get the following:

Probe Angle (degrees)	Manometer Reading (mmH2O)	Apparent Airspeed (m/s)
20	28	21.3895
10	28.5	21.5796
0	29.5	21.9549
-10	28.5	21.5796
-20	27	21.004

Table 3: Tabulated data of the pitot-static probe as it is rotated.

The apparent speeds only have a standard deviation of 0.345623 m/s, showing a very tight spread and indicating the the effects of the angling the probe are negligible, especially at small angles.

Attachment IX - Pitot-Static Probe Lateral Transverse Effects

Using the same equation as in Attachment VIII, we can find the apparent air speed across the vertical length of the testing apparatus.

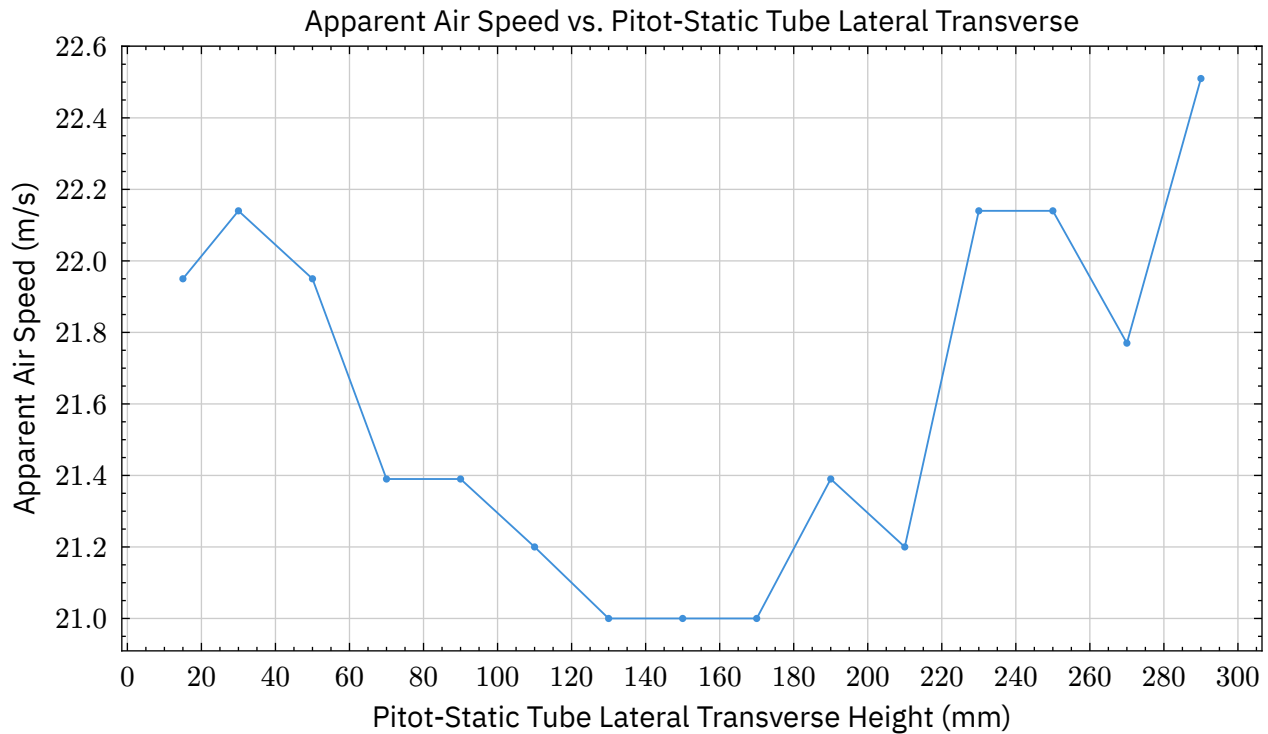


Figure 4: Plot of the calculated airspeeds as a function height of the pitot-static probe.

The resultant data shows some fluctuation in the airspeed. Specifically, as the probe crosses the center of the airflow area, we see the speed drop by over a meter per second. This implies that the velocity profile upstream of the cylinder was not uniform, likely due to the proximity of the cylinder to the entrypoint of the air. This may have contributed to some of the apparent turbulence that we see evidence of in other sections of the lab.

Attachment X - Backend Processing Code

The following was created based off of the equations provided in the lab manual. The equations were implemented in code with the help of a LLM.

```
library(units)
library(dplyr)
library(readr)
library(jsonlite)

temperature <- set_units(21, "degC")
pressure <- set_units(29.94, "mmHg")
humidity <- set_units(43, "%")
air_density <- set_units(1.200764, "kg/m^3") #adjusted for temp and humidity
u_infinity <- set_units(20, "m/s")
p_delta <- set_units(245, "Pa")
water_density <- set_units(1000, "kg/m^3")
gravity <- set_units(9.81, "m/s^2")
height_of_water <- set_units(24.97452, "mm")
# height_2 <- set_units(25, "mm")

cylinder_data <- read_csv("cylinder-rotation-data.csv", show_col_types = FALSE) %>%
  rename(
    pressure_tap_angle = `Prssure Tap Angle (degrees)`,
    cylinder_manometer = `Cylinder Manometer (kPa)`,
    airspeed_manometer = `Airspeed Manometer (mmH2O)`,
    measured_drag = `Measured Drag (N)`
  ) %>%

  mutate(
    pressure_tap_angle = set_units(pressure_tap_angle, "degree"),
    cylinder_manometer = set_units(cylinder_manometer, "kPa"),
    airspeed_manometer = set_units(airspeed_manometer - 0, "mm"),
    measured_drag = set_units(measured_drag, "N")
  )

# Convert the units for prob1
cylinder_data$cylinder_manometer <- set_units(cylinder_data$cylinder_manometer,
"Pa")
cylinder_data$airspeed_manometer <- water_density * gravity *
cylinder_data$airspeed_manometer

prob_1 <- drop_units(cylinder_data)
write_json(prob_1, "cylinder_rotation_processed.json", dataframe = "columns", pretty
= TRUE)

## Prob 2
```

```

c_p_exp <- cylinder_data$cylinder_manometer / cylinder_data$airspeed_manometer

angle_rad <- drop_units(cylinder_data$pressure_tap_angle) * pi / 180
c_p_inv <- 1 - 4 * (sin(angle_rad))^2

prob_2 <- data.frame(
  pressure_tap_angle = drop_units(cylinder_data$pressure_tap_angle),
  c_p_exp            = drop_units(c_p_exp),
  c_p_inv            = c_p_inv
)

write_json(prob_2, "c_p.json", dataframe = "columns", pretty = TRUE)

### Prob 3

cylinder_radius <- set_units(31.8, "mm")
cylinder_length <- set_units(300, "mm")

integral_sum <- set_units(0, "Pa")

for (i in 1:(nrow(cylinder_data) - 1)) {
  d_theta <- angle_rad[i + 1] - angle_rad[i]

  avg_height <- (cylinder_data$cylinder_manometer[i] * cos(angle_rad[i]) +
cylinder_data$cylinder_manometer[i + 1] * cos(angle_rad[i + 1])) / 2
  d_pressure <- avg_height * d_theta

  integral_sum <- integral_sum + d_pressure
}

print(integral_sum)
D_p = 2 * cylinder_length * cylinder_radius * integral_sum
units(D_p) <- "N"
print(D_p)

given_drag_coef <- 1.2
F_d = given_drag_coef * air_density / 2 * u_infinity ^2 * cylinder_radius ^2 *
cylinder_length # table 7-3, page 494
units(F_d) <- "N"
print(F_d)

#488
# prob 7
library(dplyr)
library(readr)
library(units)

pitot_processed <- read_csv("pitot-angle.csv", show_col_types = FALSE) %>%
  rename(
    probe_angle_deg = `Probe Angle (degrees)`,

```

```

    manometer_reading_mm = `Manometer Reading (mmH2O)`
  ) %>%
  mutate(
    # Temporarily tag columns with units for safe physics calculations
    probe_angle_deg      = set_units(probe_angle_deg, "degree"),
    manometer_reading_mm = set_units(manometer_reading_mm, "mm"),

    u_infinity           = sqrt((2 * water_density * gravity *
manometer_reading_mm) / air_density),

    probe_angle_deg      = drop_units(probe_angle_deg),
    manometer_reading_mm = drop_units(manometer_reading_mm),
    u_infinity_m_s       = signif(drop_units(u_infinity), 6)
  ) %>%
  select(-u_infinity)

u_infinity_std_dev = sd(pitot_processed$u_infinity_m_s)

write_csv(pitot_processed, "pitot_angle_processed.csv")
print(u_infinity_std_dev)
# print(pitot_processed)

pitot_lateral_processed <- read_csv("pitot-lateral.csv", check.names = FALSE) %>%
  rename(
    probe_location      = `Probe Location`,
    manometer_reading_mm = `Manometer Reading`
  ) %>%

  mutate(
    manometer_reading_mm = set_units(manometer_reading_mm, "mm"),
    u_infinity           = sqrt((2 * water_density * gravity *
manometer_reading_mm) / air_density),

    probe_location      = signif(probe_location, 4),
    manometer_reading_mm = signif(drop_units(manometer_reading_mm), 4),
    u_infinity_m_s       = signif(drop_units(u_infinity), 4)
  ) %>%
  select(-u_infinity)

write_json(pitot_lateral_processed, "pitot_lateral_processed.json", dataframe =
"columns", pretty = TRUE)

```